Ceph Performance Evaluation Report

Issei Hasegawa

2026-05-07

Table of contents

1 Overview

This report describes the setup and evaluation of a Ceph storage cluster on CloudLab. The goal of this experiment is to deploy Ceph on multiple nodes, replay a vscsi block trace workload, measure system performance, and analyze whether the results are reasonable.

2 Experimental Environment

This experiment was conducted on CloudLab using multiple nodes.

2.1 Hardware Environment

- Platform: CloudLab
- Number of nodes: TODO
- CPU: TODO
- Memory: TODO
- Storage device: TODO
- Network: TODO

2.2 Software Environment

- Operating system: TODO
- Ceph version: TODO
- Benchmark/replay tool: TODO
- Trace file: `cloudPhysicsIO.vscsi`

3 Ceph Configuration

In this experiment, a Ceph cluster was configured using several nodes.

3.1 Cluster Configuration

- MON node(s): TODO
- MGR node(s): TODO
- OSD node(s): TODO
- Client node: TODO
- Pool configuration: TODO
- Replication factor: TODO

3.2 Storage Configuration

Ceph stores data across Object Storage Daemons (OSDs). In this experiment, the available disks on the nodes were used as OSDs, and the workload was replayed from a client node against the Ceph storage system.

4 Workload

This experiment uses a vscsi block I/O trace as the workload.

4.1 Trace

The trace used in this experiment is:

```
cloudPhysicsIO.vscsi
```

This trace represents block-level I/O requests. By replaying this trace against the Ceph cluster, the experiment applies a workload that is closer to a realistic storage access pattern than a purely synthetic benchmark.

4.2 Workload Characteristics

The workload can be analyzed using the following characteristics:

- Read/write ratio: TODO
- Request size distribution: TODO
- Sequential or random access pattern: TODO
- Total number of requests: TODO
- Replay duration: TODO

5 Measurement Method

The performance of the Ceph cluster was measured while replaying the workload.

5.1 Metrics

The main metrics collected in this experiment are:

- Throughput
- IOPS
- Average latency
- Tail latency
- CPU usage
- Memory usage
- Network usage
- Disk utilization

5.2 Procedure

The experimental procedure is as follows:

1. Allocate multiple nodes on CloudLab.
2. Install and configure Ceph on the nodes.
3. Set up the Ceph cluster.
4. Prepare a storage interface such as RBD or CephFS.
5. Replay the vscsi trace workload.
6. Record performance metrics during the experiment.
7. Organize the results into tables and figures.
8. Analyze whether the observed results make sense.

6 Results

This section presents the results collected during the experiment.

6.1 Throughput

TODO: Present the throughput results using a table or figure.

6.2 IOPS

TODO: Present the IOPS results using a table or figure.

6.3 Latency

TODO: Present the average latency and tail latency results.

6.4 Resource Usage

TODO: Present CPU, memory, network, and disk utilization results.

7 Discussion

This section discusses the meaning of the experimental results.

The discussion should address questions such as:

- Was the observed throughput reasonable?
- Did the system reach any bottlenecks?
- Was latency affected by the workload characteristics?
- Did the bottleneck appear to be CPU, network, disk, or Ceph configuration?
- Did the results match the expected behavior of the vscsi trace?
- How did the Ceph configuration affect performance?

8 Limitations and Future Improvements

This experiment has several limitations.

8.1 Limitations

- The number of nodes was limited.
- The experiment may have been run only a small number of times.
- CloudLab hardware and network conditions may affect the results.
- Only one workload trace was used.
- The Ceph configuration may not have been fully optimized.
- The results may depend on the selected storage interface, such as RBD or CephFS.

8.2 Future Improvements

Future work could improve the experiment in the following ways:

- Increase the number of nodes to evaluate scalability.
- Compare different replication factors.
- Test multiple workloads, such as read-heavy and write-heavy workloads.
- Run each experiment multiple times and report averages and variation.
- Analyze latency distribution and tail latency in more detail.
- Compare different Ceph pool, OSD, and cache configurations.

9 References